Software Engineering (IT632)
Software Requirements Specification (SRS) Ver 1.0

Winter 2026

Project Title :

Smart Food Ordering and Delivery Platform

A web-based food ordering and delivery management system

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1.Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the requirements for the Smart Food Ordering and Delivery Platform.

This document defines the system's functional and non-functional requirements and serves as a reference for developers, testers, stakeholders, and instructors throughout the software development lifecycle.

1.2 Scope

The system allows:

- Customers to order food online
- Customers can schedule food orders for future delivery time
- Restaurants to manage menus and orders
- Delivery partners to receive delivery tasks
- Admin to control platform activities
- Secure online and offline payment options
- Real-time order tracking
- GPS-based delivery tracking
- Notifications for order updates
- Feedback and Ratings

The platform will operate through a web application.

1.3 Intended Audience

This document is for:

- Developers
- Designers
- Testers
- Project stakeholders
- Project team members
- Faculty evaluators

1.4 Product Features

Main features include:

- Customer registration and login
- Restaurant listing and menu viewing
- Add items to cart and place orders
- Scheduled order placement for future delivery
- Multiple payment options (online and cash on delivery)
- Order tracking
- Live GPS tracking of delivery partner
- Restaurant order management
- Delivery partner order assignment
- Delivery status updates
- Feedback and rating system
- Admin dashboard for managing users and restaurants
- Real-time notifications

1.5 Functional Requirements

MODULE 1 - User & Account Management

Requirement ID	Title	Description	Priority	Risk
FR-UM-01	User Registration	System allows customers, restaurants, and delivery partners to create accounts.	1	Critical
FR-UM-02	User Login	System authenticates users and grants role-based access.	1	Critical
FR-UM-03	Password Reset	Users can reset forgotten passwords through verification.	2	Medium
FR-UM-04	Role-Based Access	System restricts functionality according to user roles.	1	Critical
FR-UM-05	Profile Management	Users can update personal and contact details.	2	Low

MODULE 2 - Restaurant Management

Requirement ID	Title	Description	Priority	Risk
FR-RM-01	Restaurant Registration	Restaurant owners can register restaurants on the platform.	1	High
FR-RM-02	Restaurant Approval	Admin approves or rejects restaurant registrations.	1	High
FR-RM-03	Restaurant Listing	Available restaurants are displayed to customers.	1	High
FR-RM-04	Restaurant Availability	Restaurants can mark themselves available or unavailable.	2	Medium

MODULE 3 - Menu Management

Requirement ID	Title	Description	Priority	Risk
FR-MM-01	Menu Item Management	Restaurants add, update, or remove menu items.	1	High
FR-MM-02	Menu Pricing Update	Restaurants update item pricing and availability.	1	High
FR-MM-03	Menu Display	Customers can view menu details and prices.	1	High

MODULE 4 - Food Search & Discovery

Requirement ID	Title	Description	Priority	Risk
FR-FD-01	Restaurant Search	Customers search restaurants by name or cuisine.	2	Medium
FR-FD-02	Food Search	Customers search food items across restaurants.	2	Medium
FR-FD-03	Filtering & Sorting	Restaurants can be filtered or sorted.	3	Low

MODULE 5 - Cart & Order Placement

Requirement ID	Title	Description	Priority	Risk
FR-OP-01	Add to Cart	Customers add items to shopping cart.	1	Critical

FR-OP-02	Modify Cart	Customers modify or remove cart items.	1	Critical
FR-OP-03	Order Placement	Customers place immediate or scheduled orders using cart items.	1	Critical
FR-OP-04	Address Selection	Customers select or add delivery addresses.	1	High
FR-OP-05	Order Confirmation	System confirms successful orders.	1	High

MODULE 6 -Payment Management

Requirement ID	Title	Description	Priority	Risk
FR-PM-01	Payment Mode Selection	Customer selects preferred payment method (online payment or cash on delivery) before confirming the order.	1	High
FR-PM-02	Payment Processing	System processes online payments through secure gateways and supports cash on delivery orders.	1	Critical
FR-PM-03	Payment Confirmation	System confirms successful payment for online transactions and confirms order placement for cash on delivery.	1	Critical
FR-PM-04	Payment Failure Handling	Users are notified when online payment fails and are allowed to retry the transaction.	2	Medium

MODULE 7 - Order Processing (Restaurant)

Requirement ID	Title	Description	Priority	Risk
FR-OR-01	Order Notification	Restaurants receive order notifications.	1	Critical
FR-OR-02	Accept/Reject Order	Restaurants accept or reject orders.	1	Critical

FR-OR-03	Preparation Update	Restaurants update preparation status.	2	Medium

MODULE 8 - Delivery Management

Requirement ID	Title	Description	Priority	Risk
FR-DM-01	Delivery Assignment	System assigns delivery partners to orders.	1	Critical
FR-DM-02	Delivery Partner Login	Delivery partners access assigned orders.	1	High
FR-DM-03	Delivery Acceptance	Delivery partner accepts assigned order.	1	High
FR-DM-04	Delivery Status Update	Delivery progress is updated by partner.	1	High
FR-DM-05	Delivery Completion	Delivery partner marks order delivered.	1	High

MODULE 9 - Tracking & Notifications

Requirement ID	Title	Description	Priority	Risk
FR-TN-01	Order Tracking	Customers track order progress.	1	High
FR-TN-02	GPS Tracking	Customers view delivery partner location.	2	Medium
FR-TN-03	Notifications	System sends order update notifications.	2	Medium

MODULE 10-Feedback & Complaint

Requirement ID	Title	Description	Priority	Risk
FR-FC-01	Rating Submission	Customers submit ratings and reviews.	3	Low
FR-FC-02	Complaint Submission	Customers submit complaints.	2	Medium
FR-FC-03	Complaint Handling	Admin reviews and resolves complaints.	2	Medium

MODULE 11 - Admin Management

Requirement ID	Title	Description	Priority	Risk
FR-AM-01	User Management	Admin manages users and partners.	1	Critical
FR-AM-02	Restaurant Management	Admin monitors restaurants.	1	High
FR-AM-03	Order Monitoring	Admin monitors all orders.	1	High
FR-AM-04	Analytics Dashboard	Admin views performance statistics.	2	Medium

MODULE 12 -Order History & Reports

Requirement ID	Title	Description	Priority	Risk
FR-HR-01	Order History Storage	System stores customer order history.	2	Low
FR-HR-02	Order Reordering	Customers reorder past orders.	3	Low

1.6 Non-Functional Requirements

- Performance
  - GPS tracking updates should refresh smoothly.
  - The system should be available 24/7 except during scheduled maintenance.
- Security
  - Secure login and payment system.
  - User data must be protected.
- Usability
  - Easy and user-friendly interface.
- Reliability
  - System should operate with minimal downtime.
- Maintainability
  - Easy to update and improve.
- Scalability
  - Should support increasing users in the future.

1.7 Architecture Overview

Component	Technology
Frontend	Next.js / React.js
Backend	Node.js / Firebase
Database	Firebase Firestore
Authentication	Firebase Authentication
Hosting	Cloud Hosting
GPS Tracking	OpenMaps API / MapMyIndia API
Image Storage	ImageKit / Cloudinary

1.8 Future Enhancements

Possible future upgrades:

- Mobile application
- AI-based food recommendations
- Discount system
- Smart delivery route optimization
- Service of instamart quick commerce.

1.9 User Stories

ID	User Role	Wants To	So That
UR001	Customer	create an account	use the platform services
UR002	Customer	log in securely	access personal account
UR003	Customer	reset password	regain access if password is forgotten
UR004	Customer	view nearby restaurants	choose available options
UR005	Customer	view menus and prices	select food items
UR006	Customer	search food or restaurant	quickly find items
UR007	Customer	add items to cart	prepare an order
UR008	Customer	modify or remove items in cart	correct order before payment
UR009	Customer	place an order	receive food delivery
UR010	Customer	selects preferred payment method (online payment or cash on delivery)	complete the order conveniently
UR011	Customer	save delivery address	avoid entering address repeatedly
UR012	Customer	track order status	know preparation progress
UR013	Customer	track delivery using GPS	know delivery arrival time

UR014	Customer	receive notifications	stay informed about order updates
UR015	Customer	rate and review service	share experience
UR016	Customer	view past orders	reorder food easily
UR017	Restaurant Owner	register restaurant	receive online orders
UR018	Restaurant Owner	manage menu items	update food availability
UR019	Restaurant Owner	accept or reject orders	manage workload
UR020	Restaurant Owner	update order preparation status	inform customers
UR021	Delivery Partner	log in to system	access delivery assignments
UR022	Delivery Partner	accept delivery requests	deliver orders
UR023	Delivery Partner	update delivery status	inform system about delivery progress
UR024	Delivery Partner	share live GPS location	enable tracking for customers
UR025	Admin	manage users and restaurants	maintain platform operations
UR026	Admin	monitor all orders	ensure smooth system operation
UR027	Admin	handle complaints and issues	maintain service quality

1.10 Project Plan

Overview

• Duration: February – April 2026 (12 weeks)
• Methodology: Agile with weekly sprints and continuous feedback

The project will be executed using an iterative Agile approach where each phase is divided into weekly sprints. Each sprint focuses on completing a specific set of tasks, followed by review and improvements. This approach ensures steady progress, early issue detection, and high-quality documentation.

Phase 1: Requirement Analysis & Planning (Week 1–2)

Deliverables:

• Finalized Software Requirements Specification (SRS)
• Functional and non-functional requirement validation
• Use case diagrams and user story refinement
• Feature prioritization for development sprints

Activities:

• Stakeholder discussions
• Understanding food ordering and Instamart-like grocery flow
• Requirement documentation and approval

Phase 2: System Design (Week 3–4)

Deliverables:

• UI wireframes for customer, restaurant, delivery partner, and admin panels
• Database schema design using Firebase Firestore

• System architecture diagrams
• User flow and workflow diagrams

Activities:

• Designing responsive interfaces
• Structuring real-time database relationships
• Defining API interactions

Phase 3: Backend Development (Week 5–7)

Deliverables:

• Firebase database implementation
• Node.js backend APIs
• User authentication and role management
• Order processing system
• Instamart product management module
• GPS real-time tracking services

Activities:

• API development
• Payment workflow integration

• Real-time order updates
• Location data handling

Phase 4: Frontend Development (Week 8–9)

Deliverables:

• Customer food ordering interface
• Instamart grocery ordering interface
• Restaurant order management dashboard

• Admin control panel
• Live map tracking interface

Activities:

• UI development in React/Next.js
• Connecting frontend with backend APIs
• Implementing cart and checkout flows
• Integrating Google Maps services

Phase 5: Testing & Quality Assurance (Week 10)

Deliverables:

• Functional testing reports
• GPS accuracy testing
• Payment testing
• Bug resolution logs

Activities:

• Manual testing of all modules
• Performance checks
• Fixing identified issues

Phase 6: Deployment & Documentation (Week 11)

Deliverables:

• Cloud-hosted application
• Database production setup
• Final project documentation
• User manuals

Activities:

• Deploying frontend and backend
• Configuring Firebase services
• Preparing academic documentation

Phase 7: Final Submission & Launch (Week 12)

Deliverables:

• Fully working system
• Project report
• Presentation material
• User onboarding demo

Activities:

• Final testing pass
• System walkthrough
• Evaluation preparation

1.11 Tools And Technologies to be used

Technology	Layer Used	Reason
Next.js / React.js	Frontend	Provides fast and responsive interfaces for browsing food items, Instamart groceries, and placing orders
Node.js	Backend	Handles server-side processing, APIs, and business logic
Firebase	Backend Services	Manages real-time updates, notifications, and cloud-based services
Firebase Firestore	Database	Stores user data, orders, food items, and grocery products with real-time synchronization
Firebase Authentication	Backend	Provides secure login and role-based user access
OpenMaps API / MapMyIndia API	Backend / Frontend	Enables live GPS tracking and delivery route visualization
Jira, GitHub	Project Management	Supports sprint planning, task tracking, and version control

1.12 Timeline Overview / Sprints

Phase	Duration (Weeks)	Sprint(s)
Requirement Analysis & Planning	2	Sprint 1–2
Design	2	Sprint 3–4
Backend Development (Node.js + Firebase)	3	Sprint 5–7
Frontend Development	2	Sprint 8–9
Testing	1	Sprint 10
Deployment & Documentation	1	Sprint 11
Final Submission & Review	1	Sprint 12

1.13 Gantt Chart / Project Plan

Week	Requirement Analysis & Planning	Design	Backend Development	Frontend Development	Testing	Deployment & Documentation	Final Submission & Review
Week 1	■						
Week 2	■						
Week 3		■					
Week 4		■					
Week 5			■				
Week 6			■				
Week 7			■				
Week 8				■			
Week 9				■			
Week 10					■		
Week 11						■	
Week 12							■